CAMPUS CAFE
ORDERING SYSTEM

Java Object-Oriented Programming Laboratory Exercise

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Course / Section:

BSINT – 2 ITS202 - OBJECT-ORIENTED PROGRAMMING/SECTION – D

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1. OOP Concepts and Their Applications

Abstraction – We used the abstract class MenuItem to define common properties and the computePrice() method. This allows each menu item to have its own pricing calculation.

Encapsulation – We used private attributes and getters and setters to protect the data. Setters also validate values, such as preventing negative prices.

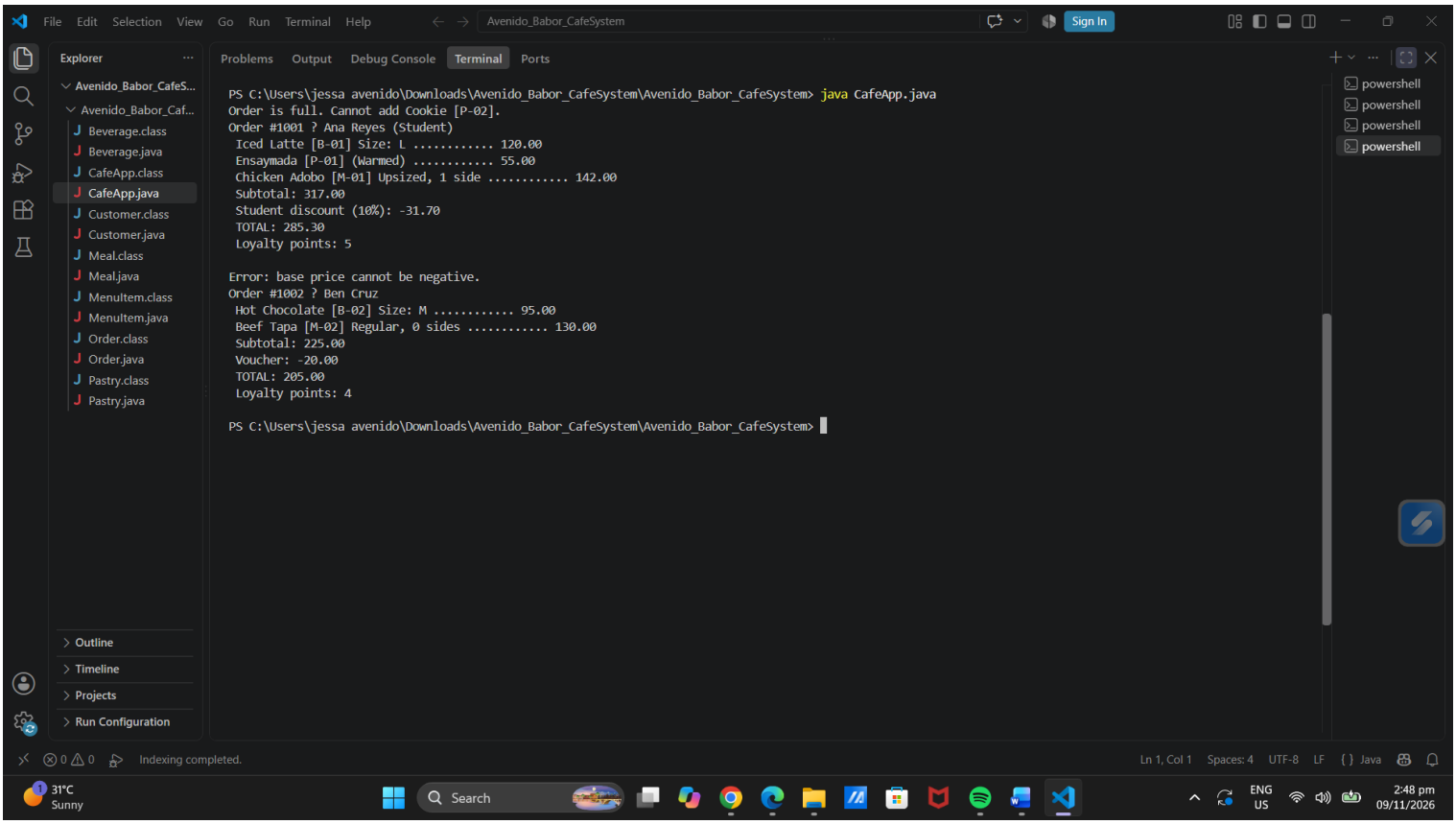
Inheritance – The classes Beverage, Pastry, and Meal inherit from the parent class MenuItem. This allows them to reuse common attributes and methods.

Polymorphism – We used polymorphism by storing different menu items using the MenuItem type. Each item performs its own overridden pricing and description methods.

Method Overloading – We used overloaded constructors and methods in Customer and Order, allowing the same method name to work with different parameters.

Method Overriding – The Beverage, Pastry, and Meal classes override computePrice() and getDescription() to provide their own implementations.

2. Actual Program Output



3. Challenges Encountered and Solutions

One challenge we encountered was implementing different pricing rules for beverages, pastries, and meals. We solved this by overriding the computePrice() method in each subclass.

Another challenge was preventing invalid values, such as negative prices. We used conditional statements in setter methods to validate the input.

We also needed to limit each order to three items. We solved this by using three separate item fields and checking the item count before adding another item.

Lastly, we implemented student discounts and vouchers using overloaded methods to calculate the correct order totals.

4. Conclusion

Through this activity, we gained a better understanding of Java Object-Oriented Programming. We learned how to apply abstraction, encapsulation, inheritance, polymorphism, method overloading, and method overriding in creating a functional Café Ordering System.